

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section:

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.

- The Delta tables for customer and order data do not exist or are empty.

Main Flow / Functional Steps:

- Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into a Delta table named "customer".
- Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into a Delta table named "order".

The schema for the "customer" table is: CustId, Name, EmailId, Region.

The schema for the "order" table is: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.

FR-DLT-002: Transform Order Data by Adding TotalAmount Column

Title: Transform Order Data by Adding TotalAmount Column

Description: The system must add a new column "TotalAmount" to the "order" Delta table, which is the product of "PricePerUnit" and "Qty" columns.

Preconditions:

- The "order" Delta table exists and is populated with data.

Main Flow / Functional Steps:

- Add a new column "TotalAmount" to the "order" Delta table.
- Populate the "TotalAmount" column by multiplying the "PricePerUnit" and "Qty" columns.

FR-DLT-003: Cleanse Customer and Order Data

Title: Cleanse Customer and Order Data

Description: The system must remove null and duplicate records from both "customer" and "order" Delta tables.

Preconditions:

- Both "customer" and "order" Delta tables exist and are populated with data.

Main Flow / Functional Steps:

- Remove null records from both "customer" and "order" Delta tables.
- Remove duplicate records from both "customer" and "order" Delta tables.

FR-DLT-004: Create ordersummary Table and Load Joined Data

Title: Create ordersummary Table and Load Joined Data

Description: The system must create an "ordersummary" table if it does not exist and load joined data from "customer" and "order" Delta tables using the "CustId" field.

Preconditions:

- Both "customer" and "order" Delta tables exist and are populated with cleansed data.
- The "ordersummary" table does not exist or is empty.

Main Flow / Functional Steps:

- Create the "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- Join the "customer" and "order" Delta tables using the "CustId" field.
- Load the joined data into the "ordersummary" table, which is an SCD type 2 table.

FR-DLT-005: Implement SCD Type 2 Logic for ordersummary Table

Title: Implement SCD Type 2 Logic for ordersummary Table

Description: The system must update the "ordersummary" SCD type 2 table whenever there is a change in the "customer" table.

Preconditions:

- The "ordersummary" table exists and is populated with data.
- The "customer" table exists and is populated with data.

Main Flow / Functional Steps:

- When a change is detected in the "customer" table, update the corresponding records in the "ordersummary" table.
- Make old records inactive and new records active in the "ordersummary" table.
- Update the StartDate and EndDate columns accordingly to maintain history.

FR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data**Title: Create customeraggregatespend Table and Load Aggregated Data**

Description: The system must create a "customeraggregatespend" table if it does not exist and load aggregated data from the "ordersummary" table.

Preconditions:

- The "ordersummary" table exists and is populated with data.
- The "customeraggregatespend" table does not exist or is empty.

Main Flow / Functional Steps:

- Create the "customeraggregatespend" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdic_wizard".
- Aggregate the "TotalAmount" column from the "ordersummary" table, grouping by "Name" and "Date" columns.
- Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-007: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement Delta Live Tables to execute the above steps.

Preconditions:

- All previous steps have been executed successfully.

Main Flow / Functional Steps:

- Implement Delta Live Tables to automate the execution of the above steps.